

# Splink Cohort Builder

## Linkage Analysis Report

Run 1

Generated: 2026-07-06 09:06

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### Run Configuration

|                    |                                                         |
|--------------------|---------------------------------------------------------|
| Operation mode:    | Link Dedupe                                             |
| Linkage type:      | Probabilistic                                           |
| Fields used:       | first_name, surname, dob, city, email, gender, postcode |
| Active blocking:   | first_name, surname, dob, city, email, gender, postcode |
| Cluster threshold: | 0.8                                                     |

## 1. Dataset Overview

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This run performed linkage between Dataset A (fake1000, 1000 records) and Dataset B (50% sample of Dataset A with controlled errors introduced). Dataset B contains: 14% first-name typos, 9% surname typos, 5% missing DOBs, 15% email variations, 11% city abbreviations, and 7% gender assignment errors. The goal was to identify matching records across the two datasets.

**Total records processed:** 1500

## 2. Blocking Rules

Blocking rules define which record pairs are brought forward for comparison. Two records are only compared if they match exactly on at least one blocking field. This reduces the  $O(n^2)$  comparison space to a tractable number of candidate pairs. Turning off blocking rules increases recall but also increases computation time.

| Field      | Blocking Rule Active |
|------------|----------------------|
| first_name | Yes                  |
| surname    | Yes                  |
| dob        | Yes                  |
| city       | Yes                  |
| email      | Yes                  |
| gender     | Yes                  |
| postcode   | Yes                  |

### 3. Edge Metrics

An 'edge' is a predicted pairwise link between two records. For probabilistic linkage, each edge has a match\_probability between 0 and 1. Values close to 1.0 indicate near-certain matches; values near 0.5 are uncertain. For deterministic linkage, all edges are assigned match\_probability = 1.0.

#### Edge Counts

Number of predicted edges: 2475

Unique IDs with at least one edge: 1411

#### Match Probability Statistics

| Statistic | Value  |
|-----------|--------|
| Mean      | 0.7474 |
| Median    | 0.9947 |
| Min       | 0.0307 |
| Max       | 1.0    |
| Std Dev   | 0.3713 |

Interpretation: A high mean match\_probability (>0.9) suggests the model is confident in its predictions. A wide standard deviation indicates a mix of high- and low-confidence edges - inspect lower-confidence edges carefully.

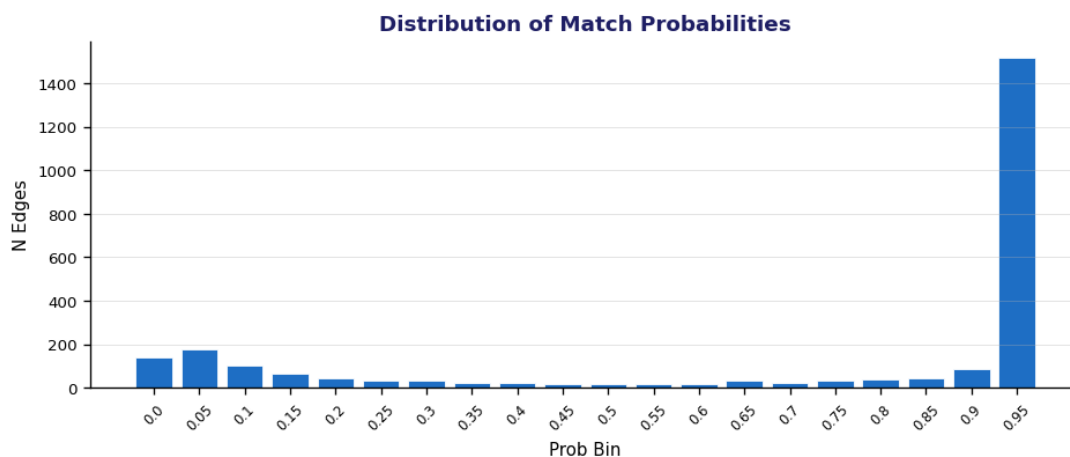


Figure: Number of edges at each match probability band. High bars near 1.0 indicate strong model confidence.

#### Gamma Score Averages

Gamma scores measure agreement level between record pairs for each comparison field. gamma=1 means exact agreement; gamma=0 means total disagreement; intermediate values represent partial matches (e.g. Jaro-Winkler fuzzy similarity). Higher mean gammas indicate better overall agreement on that field.

| Field      | Mean Gamma Score |
|------------|------------------|
| first_name | 1.9511           |
| surname    | 1.8101           |

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|          |         |
|----------|---------|
| dob      | 3.0586  |
| city     | 0.1479  |
| email    | -0.0315 |
| gender   | 0.6937  |
| postcode | -0.3511 |

## 4. Cluster Metrics

After pairwise prediction, connected components clustering groups records into entity clusters. Each cluster ideally represents one real-world individual. Singletons (clusters of size 1) are records with no predicted matches. Larger clusters may indicate correct multi-record entities or over-linking errors.

### Cluster Counts

**Total distinct clusters:** 437

**Cross-dataset clusters:** 261

### Singleton vs Multi-Record Clusters

| cluster_type                      | n_clusters | total_records |
|-----------------------------------|------------|---------------|
| Multi-record cluster (2+ records) | 261        | 1324.0        |
| Singleton (1 record)              | 176        | 176.0         |

Interpretation: A high singleton count means many records could not be linked. This may be acceptable (genuinely unique records) or may indicate blocking rules that are too restrictive. Multi-record clusters represent found duplicates.

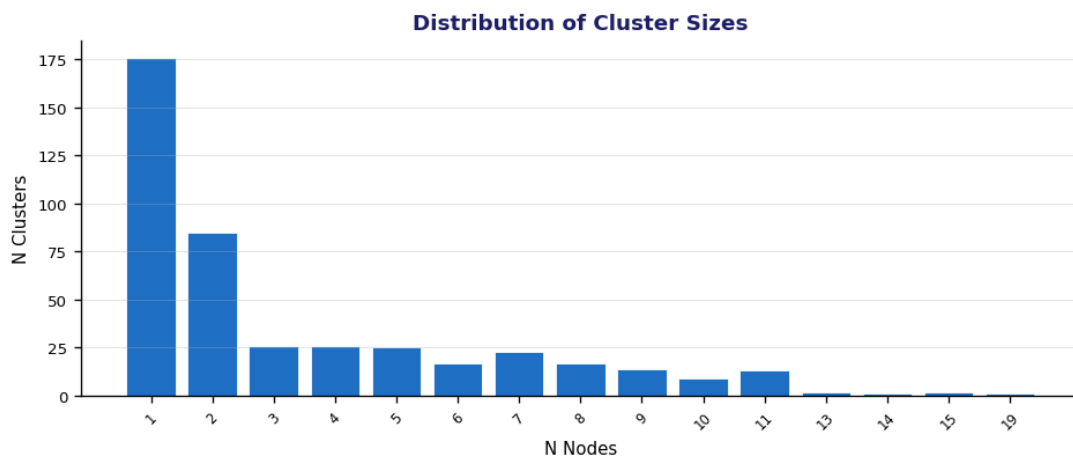


Figure: How many clusters contain each number of records. A J-shaped curve (many singletons, few large clusters) is typical for real data.

### Gender Distribution in Linked Records

| gender  | n_records | pct  |
|---------|-----------|------|
| M       | 718       | 47.9 |
| F       | 532       | 35.5 |
| Unknown | 250       | 16.7 |